Arun Sai Thunga

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Summary

AI Engineer specializing in Large Language Models (LLMs), retrieval-augmented generation (RAG), and machine learning systems. Experienced in building LLM-driven backend services, RAG pipelines, and ML workflows using Python, FastAPI, Docker, and Kubernetes across Linux and cloud environments (Azure, AWS). Hands-on experience in LLM fine-tuning (LoRA/QLoRA), prompt engineering, hallucination evaluation, and data pipeline development through roles at Calvergy, Deeptronics, and FAPS Nürnberg. Strong background in ML model development, data engineering, and scalable AI system deployment. German B2, English C1.

Software Knowledge

Programming & Systems: Python, C++, Java, Linux, Git, Shell Scripting
LLMs & NLP: Large Language Models (LLMs), RAG Pipelines, Prompt Engineering, Transformers, Hugging Face, LangChain, LLaMA, Mistral, LoRA / QLoRA
Machine Learning: PyTorch, scikit-learn, Synthetic Data Generation, Model Evaluation, Latency & Hallucination Analysis
MLOps & DevOps: Docker, Kubernetes, CI/CD, MLflow, Model Versioning, Automated Deployment
Cloud Platforms: Microsoft Azure, AWS, Azure ML, AKS
Development Tools & Visualization: Eclipse, VS Code, Jupyter Notebooks, Power BI, Grafana, Prometheus

Professional Experience

AI-Backend Developer Intern — 10/2025 – 01/2026
Calvergy , Aachen, Germany

- Designed and integrated LLM-driven backend services using Python, FastAPI, and Java-compatible interfaces, operating in Linux environments with command-line workflows for industrial energy platforms.
- Implemented RAG-based pipelines with document parsing, chunking strategies, and prompt engineering, improving system explainability, response correctness, and decision support accuracy.
- Curated and pre-processed structured and unstructured datasets using Pandas and NumPy, supporting LLM evaluation on correctness, latency, and hallucination metrics.

DevOps & Machine Learning Intern — 11/2024 –03/2025
Deeptronics GmbH, Lauf, Germany

- Developed LLM-ready data pipelines with advanced preprocessing, statistical analysis, and chunking strategies, enabling efficient ingestion into RAG systems.
- Built CI/CD workflows for LLM-based services using Docker, Kubernetes, and shell scripting, supporting containerized deployment, automated testing, and controlled rollouts.
- Supported MLOps pipelines for model versioning, evaluation, and retraining using MLflow and Git, ensuring reproducibility and stable production operation.

Student Research Assistant in Machine Learning and AI — 08/2022 –10/2024
Chair of Factory Automation and Production Systems(FAPS), Nürnberg, Germany

- Designed and deployed LLM-powered inference services on Azure AKS using Docker and Kubernetes, optimizing response latency, scalability, and reliability for research prototypes.
- Fine-tuned open-source transformer models using PyTorch and Hugging Face, applying LoRA and QLoRA techniques for parameter-efficient training.
- Evaluated LLM systems using quantitative metrics, latency benchmarks (<300 ms), and hallucination analysis.
- Developed data analysis and visualization dashboards using Power BI, enabling model performance comparison and A/B testing on data from PostgreSQL and cloud-based sources.

Master's Thesis – Anomaly Detection in Time Series using ML/DL 11/2023 – 06/2024
AVL List GmbH, Graz, Austria

- Developed ETL pipelines with advanced preprocessing techniques to handle class imbalance, successfully improving model training accuracy to 95%.
- Fine-tuned high-performance multi-output ML/DL models using PyTorch and TensorFlow, achieving 98.7% F1-score and enabling real-time predictions through Simulink integration.

Relocation to Germany & preparation for M.Sc. studies 04/2021 – 03/2022

- Completed advanced coursework in C++ and machine learning algorithms, strengthening foundations in data structures, optimization, and statistical learning.

Junior Data Scientist 06/2019 – 03/2021
Decision Point Analytics, Delhi, India

- Optimized ETL pipelines using Python, Apache Spark, and SQL to streamline data ingestion and processing across cloud-based data sources.
- Developed and deployed machine learning models (regression, decision trees, ensemble methods) using Amazon SageMaker to support data-driven product and analytics use cases.
- Automated data validation and quality checks within ETL workflows for SQL and NoSQL data sources, improving data consistency and model reliability.

Education

M.Sc. in Medical Engineering, specialization in AI and Data Science *04/2022 – 03/2026*
Friedrich-Alexander-Universität Erlangen-Nürnberg, Germany

B.Tech. in Instrumentation Engineering *09/2015 – 05/2019*
Sree Vidyanikethan Engineering College, Tirupathi, India

Recent Projects

LLM Customer Support Agent (LangGraph + RAG + Kafka)

- Built a multilingual LLM agent using LangGraph and Hugging Face Transformers with RAG and vector search for contextual response generation.
- Implemented an event-driven backend with FastAPI and Kafka and deployed with Docker/Kubernetes, reducing response latency by ~33% and enabling scalable asynchronous inference.

Certifications

- *Microsoft Azure AI Engineer Associate* – ***Microsoft***
- *Data Science Workshop* – ***Snowflake***
- *Terraform Associate* – ***HashiCorp***
- *Machine Learning Algorithms* – ***Coursera***
- *Microsoft Azure DevOps Engineer Expert* – ***Microsoft (Pursuing)***

Languages

English: C1 (Business fluent)
German: B2(Intermediate)
Telugu: Native

Availability: Immediate start
Work Authorization: Authorized to work in Germany